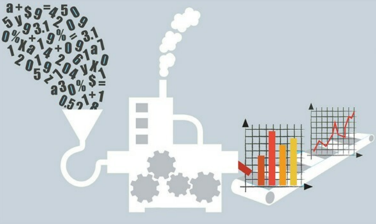
PolSARpro: Wonderful FOSS for Synthetic Aperture Radar Data Processing

Synthetic Aperture Radar (SAR) is used to create 2D or 3D images of an object. Processing of SAR data can help to show how such imagery is formed. This article on the free and open source PolSARpro will interest students, researchers, and those interested in becoming remote sensing specialists or SAR system designers.



Developed under a contract with ESA (European Space Agency)-ESRIN (ESA Centre for Space Observation), PolSARpro is a polarimetric Synthetic Aperture Radar (SAR) data processing and education tool. Most of the SAR sensors provide data at two different levels. Level 1 is complex raw data and needs a lot of mathematical and physical understanding of the phenomenon before it can be used. Level 2, on the other hand, is calibrated, geo-referenced data and a majority of the processing is image processing. In this article, we demonstrate Level 1 SAR processing through FOSS PolSARpro ver 5.1.

PolSARpro supports data processing of both airborne and space-borne radars. Table 1 lists the airborne and space-borne satellites launched by the national space exploration programmes of various countries.

Table 1

Airborne satellites	Space-borne satellites
• Airsar	• Alos-1-Palsar
• Convair	• Alos-2-Palsar
• Emisar	• Cosmo-Skymed
• E-Sar	• Envisat-Asar
• F-Sar	• Radarsat2
• Sethi	• Risat
• Uavsar	• Sentinel-1
	• SIR-C
	• TerraSAR X

Among the above mentioned satellites, data of Sentinel-1a, Sentinel-1b, Uavsar and Alos Palsar is freely available and can be found from the link *https://vertex.daac.asf.alaska.edu/?*

Processing capabilities of PolSARpro

PolSARpro is packed with tools for the smoothening of images and for speckle removal. Many filters, such as An-Yang filter, Box-car filter, Box-car Edge filter, Gaussian filter, IDAN filter, Lee Refined filter, Lee Sigma filter, Lopex filter, Mean-Shift filter, Non-local Means filter, etc, are available at your fingertips while working with this FOSS solution.

For visual appeal, you will find the Refined Lee filter applied on a SAR image in Figure 2.

There are significant decomposition algorithms available for quadrature polarimetric data, dual polarimetric data as well as compact polarimetric data. These decomposition algorithms help the study of particular features like ship detection, oil spill detection, agricultural applications, etc. A rich set of popular decomposition algorithms is available with PolSARpro. A few of them are listed in Table 2.

A step-by-step decomposition of SAR data through PolSARpro version 5.1 is shown via the home screen of this version.

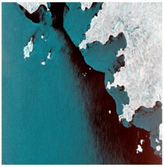


Figure 1: Image without speckle filtering

Figure 2: Image with speckle filtering